Varun Sharma

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Education

Bennett University — Greater Noida, UP
Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence) — *Expected May 2028*

- CGPA: 7.4/10.0
- Relevant Coursework: Data Structures & Algorithms, AI, Machine Learning, Linear Algebra, OOP

Technical Skills

Programming Languages: Python (Proficient), C++ (DSA), TypeScript, JavaScript
Web & Backend: FastAPI, React, SQLAlchemy, Alembic, MongoDB, REST APIs, Google OAuth, Razorpay API
Data Science & AI/ML: NumPy, Pandas, Scikit-learn, Matplotlib, Google Gemini API
Developer Tools: Git, GitHub, GitHub Actions, pytest, Vercel, VS Code, Jupyter Notebooks
Core Competencies: Data Wrangling, Time-Series Analysis, Predictive Modeling, Backtesting, Risk Analytics

Technical Projects

Freight Invoice Auditing Agent | *Python, FastAPI, Pydantic, Gemini API, React, TypeScript* — 2026
Live demo · Code · Solo project

- An LLM extracts carrier invoices; a deterministic rule engine audits every line for 8 types of overcharge.
- Built a labelled synthetic test set (5 invoice layouts, one held out) and ran an ablation: letting the LLM judge gave 65.7% precision and USD 7,030 in false claims vs 98.1% and USD 138 for the hybrid.
- Shipped reviewer sign-in, approval-gated disputes, a hash-chained audit log and 97 tests in CI; deployed on Vercel.

Algo Trade Simulator | *Python, FastAPI, React, TypeScript, MongoDB, Gemini API* — 2026
Live demo · Code · Team project with Yashika Garg

- Built a full-stack backtesting platform that simulates trades on real market data with fees, slippage and no look-ahead bias, benchmarked against buy & hold and the S&P 500 on Sharpe, Sortino, drawdown and beta.
- Implemented walk-forward testing (tune on the past year, trade the next quarter on unseen data), which showed a tuned strategy's return falling from 24% to 11% a year out of sample.
- Developed a logistic-regression trading strategy from scratch with time-aware retraining, evaluated against a majority-class baseline and ROC-AUC.
- Integrated Gemini function calling so the AI copilot runs backtests and simulations; deployed on Vercel with Google OAuth and 107 automated tests in CI.

RazorGrowth: Permissioned AI Revenue-Recovery Agent | *Python, FastAPI, React, TypeScript* — 2026
Live demo · Code · Razorpay AI Buildathon 2026 · Solo project

- Built a Gemini tool-calling agent that investigates failed payments with read-only tools and proposes Razorpay recovery campaigns; every proposal must pass schema validation, a deterministic policy engine and explicit human approval.
- Engineered safe execution: atomic approvals, deterministic Razorpay reference IDs with retry, backoff and safe halt (no duplicate links), HMAC-verified webhooks and an audit trail.
- Built an evaluation harness on a deterministic market simulator; it exposed a bug in the agent's outcome estimates, and after the fix the agent beats a heuristic baseline by 2.3%.
- Shipped Google sign-in, Alembic migrations and 70 automated tests in CI; deployed on Vercel and FastAPI Cloud.

AgriGuard: Satellite Imagery Crop Analysis | *Python, Scikit-learn, Sentinel-2 Data* — 2026

- Developed a classification system to identify crop types and detect diseases using multispectral imagery.
- Implemented image preprocessing and feature extraction on large-scale Kaggle datasets.

Leadership & Experience

Management Lead · *Sportikon 3.0* — 2025

Peer Mentor · *DSA Summit* — 2025

Honors & Activities

- **Open Source:** Maintaining public repos for AI/ML and algorithmic research.
- **Hackathons:** Participant, Razorpay AI Buildathon 2026 (built RazorGrowth).